

Mitja Devetak

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Profile

Applied mathematician focused on simulation-heavy economic problems. Specializes in agent-based modelling, unstructured data, and network analysis. Most interested in difficult, open-ended problems in finance and economics where the solution is not obvious. Experienced in turning academic research into practical tools for private-sector work and public-sector policy analysis.

Education

Complexity Science Hub Vienna

PhD in Complex Systems Science

2024-present

University of Oxford

MSc Mathematical Modelling and Scientific Computing, Distinction

2022-2023

University of Groningen

BSc Mathematics, cum laude; BA Political Philosophy, cum laude

2019-2022

Experience

Supply Chain Intelligence Institute Austria (ASCII)

Researcher and PhD Candidate

2024-present

- Built TIDES, a large-scale (40k agents, 2k ports) ABM to estimate the economic impacts of maritime disruptions.
- Translated research into ministry briefings and policy reports.
- Used LLMs to extract supply-chain information from company websites.

Freelance Quantitative Developer

Various

2024-present

- Built and deployed Python/JAX natural-disaster ABM workflows on AWS EC2.
- Developed C++ valuation modules for climate stress tests.

Université Paris 1 Panthéon-Sorbonne

Research Assistant under Antoine Mandel

2023-2024

- Co-developed BeforeIT.jl for high-performance macroeconomic ABM.

Skills

Data science: statistical modelling, large-scale data analysis, network modelling, simulation, decision-support analytics

Programming: Julia, C/C++, Python/JAX; Docker, AWS EC2, Git, Linux/Vim, LaTeX

Methods: agent-based modelling, stochastic simulation, Markov models, scenario analysis

Additional

Passed CFA Level I. RWRI 18: Risk Taking.